

AERIS Expert Review Packet

Anomaly

Source / metric	tceq / Carbon monoxide (co)
Observed / expected baseline	0.4 / 0.127607 ppm
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-06-21 03:00 UTC
Location	29.7337, -95.2576
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	9 / 610	59.83 to 100; mean 85.6873	94.16 at 2026-06-21 02:55Z; 5 min before event
asos	Air temperature (temperature)	degC	9 / 610	23 to 33.8889; mean 27.5848	24 at 2026-06-21 02:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	9 / 610	0 to 360; mean 131.377	60 at 2026-06-21 02:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	9 / 610	0 to 9.77444; mean 3.2528	2.05778 at 2026-06-21 02:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	10 / 120	5875.31 to 5915.93; mean 5894.91	5881.55 at 2026-06-21 00:00Z; 3 h before event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	10 / 120	49.5963 to 1845; mean 560.016	212.459 at 2026-06-21 00:00Z; 3 h before event
noaa_gfs	Precipitable water (precipitable_water)	mm	10 / 120	41.18 to 60.3207; mean 53.1083	53.8544 at 2026-06-21 00:00Z; 3 h before event
noaa_gfs	Surface pressure (surface_pressure)	hPa	10 / 120	1004.61 to 1012.77; mean 1009.2	1008.42 at 2026-06-21 00:00Z; 3 h before event
noaa_gfs	850-hPa temperature (t_850)	degC	10 / 120	18.2549 to 22.3564; mean 19.494	21.7464 at 2026-06-21 00:00Z; 3 h before event
noaa_gfs	10-m eastward wind (u_10m)	m/s	10 / 120	-3.27924 to 1.55084; mean -1.0036	-1.96027 at 2026-06-21 00:00Z; 3 h before event
noaa_gfs	10-m northward wind (v_10m)	m/s	10 / 120	-1.84918 to 6.63198; mean 3.0634	-0.369185 at 2026-06-21 00:00Z; 3 h before event
openaq	Ozone (ozone)	ppm	16 / 1028	-0.002 to 0.04; mean 0.0158	0 at 2026-06-21 03:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 217	6 to 95; mean 29.5622	21 at 2026-06-21 03:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	82 / 2864	0 to 93.73; mean 6.5939	6.6 at 2026-06-21 03:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 360	0 to 100; mean 64.9194	100 at 2026-06-21 03:00Z; at event time
openweather	Relative humidity (humidity)	percent	5 / 360	58 to 97; mean 85.95	95 at 2026-06-21 03:00Z; at event time
openweather	Precipitation rate (precipitation)	mm/h	5 / 360	0 to 4.74; mean 0.3589	0 at 2026-06-21 03:00Z; at event time
openweather	Surface pressure (pressure)	hPa	5 / 360	1009 to 1015; mean 1011.59	1011 at 2026-06-21 03:00Z; at event time
openweather	Air temperature (temperature)	degC	5 / 360	23.58 to 34.02; mean 28.0261	24.16 at 2026-06-21 03:00Z; at event time
openweather	Wind direction (wind_direction)	degree	5 / 360	0 to 355; mean 157.439	85 at 2026-06-21 03:00Z; at event time
openweather	Wind speed (wind_speed)	m/s	5 / 360	0.45 to 5.64; mean 2.3762	0.89 at 2026-06-21 03:00Z; at event time
purpleair	PM2.5 (pm25)	ug/m3	66 / 4703	0 to 359.8; mean 8.3517	8.1 at 2026-06-21 03:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-20 19:11Z; 7 h 49 min before event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-20 19:11Z; 7 h 49 min before event
sentinel5p	CO column (s5p_co_column)	mol/m^2	4 / 4	0.0278085 to 0.0279886; mean 0.0279	0.0279886 at 2026-06-21 18:26Z; 15 h 27 min after event
sentinel5p	CO granules available (s5p_co_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-20 19:11Z; 7 h 49 min before event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	5 / 5	0.000105409 to 0.000168978; mean 0.0001	0.000168978 at 2026-06-21 18:56Z; 15 h 56 min after event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-06-20 19:11Z; 7 h 49 min before event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	3 / 3	1.683e-05 to 5.0006e-05; mean 0	3.59018e-05 at 2026-06-21 18:56Z; 15 h 56 min after event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	4 / 4	1 to 1; mean 1	1 at 2026-06-20 19:11Z; 7 h 49 min before event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-20 19:11Z; 7 h 49 min before event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	5 / 5	-0.00020099 to -4.39631e-06; mean -0.0001	-4.39631e-06 at 2026-06-21 18:56Z; 15 h 56 min after event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-06-20 19:11Z; 7 h 49 min before event
tceq	Carbon monoxide (co)	ppm	3 / 212	0 to 0.8; mean 0.2434	0.4 at 2026-06-21 03:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	12 / 813	-2.4 to 24.5; mean 4.5969	17.3 at 2026-06-21 03:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 209	-1.4 to 13.5; mean -0.1608	-0.1 at 2026-06-21 03:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	06-19 12Z 67.73, 18Z 73.87, 06-20 00Z 87.39, 06Z 92.31, 12Z 84.14, 18Z 91.64, 06-21 00Z 93.71, 06Z 96.51, 12Z 88.39, 18Z 71.07, 06-22 00Z 85.81, 06Z 91.7, 12Z 81.13
asos	Air temperature (temperature)	06-19 12Z 31.99, 18Z 30.87, 06-20 00Z 27.36, 06Z 26.82, 12Z 28.28, 18Z 24.66, 06-21 00Z 24.52, 06Z 24.93, 12Z 25.89, 18Z 30.87, 06-22 00Z 27.96, 06Z 27.3, 12Z 29.46
asos	Wind direction (wind_direction)	06-19 12Z 174.4, 18Z 170.9, 06-20 00Z 123.1, 06Z 119, 12Z 60.21, 18Z 124, 06-21 00Z 50.83, 06Z 84.2, 12Z 161.7, 18Z 166.6, 06-22 00Z 155.4, 06Z 164.8, 12Z 175.2
asos	Wind speed (wind_speed)	06-19 12Z 4.363, 18Z 3.611, 06-20 00Z 3.719, 06Z 2.486, 12Z 2.111, 18Z 2.851, 06-21 00Z 1.093, 06Z 1.883, 12Z 3.789, 18Z 5.504, 06-22 00Z 3.925, 06Z 3.23, 12Z 4.344
noaa_gfs	500-hPa geopotential height (gh_500)	06-19 18Z 5910, 06-20 00Z 5900, 06Z 5903, 12Z 5881, 18Z 5896, 06-21 00Z 5882, 06Z 5885, 12Z 5876, 18Z 5890, 06-22 00Z 5894, 06Z 5915, 12Z 5908
noaa_gfs	Planetary boundary layer height (pbl_height)	06-19 18Z 1600, 06-20 00Z 200.7, 06Z 530.1, 12Z 240.9, 18Z 927.8, 06-21 00Z 186.6, 06Z 140.2, 12Z 299, 18Z 616.7, 06-22 00Z 879, 06Z 552.2, 12Z 547
noaa_gfs	Precipitable water (precipitable_water)	06-19 18Z 51.71, 06-20 00Z 59.04, 06Z 55.09, 12Z 52.71, 18Z 55.49, 06-21 00Z 53.48, 06Z 56.2, 12Z 57.95, 18Z 56.17, 06-22 00Z 53.65, 06Z 43.63, 12Z 42.18
noaa_gfs	Surface pressure (surface_pressure)	06-19 18Z 1010, 06-20 00Z 1009, 06Z 1010, 12Z 1010, 18Z 1011, 06-21 00Z 1008, 06Z 1009, 12Z 1008, 18Z 1009, 06-22 00Z 1007, 06Z 1009, 12Z 1010
noaa_gfs	850-hPa temperature (t_850)	06-19 18Z 18.81, 06-20 00Z 19.89, 06Z 19.91, 12Z 18.56, 18Z 18.73, 06-21 00Z 21.45, 06Z 19.48, 12Z 18.62, 18Z 19.51, 06-22 00Z 20.18, 06Z 19.92, 12Z 18.85
noaa_gfs	10-m eastward wind (u_10m)	06-19 18Z -0.3473, 06-20 00Z -1.751, 06Z -1.474, 12Z -0.782, 18Z -2.906, 06-21 00Z -1.932, 06Z -0.783, 12Z 0.4488, 18Z -0.4373, 06-22 00Z -0.9744, 06Z -0.617, 12Z -0.4874
noaa_gfs	10-m northward wind (v_10m)	06-19 18Z 4.553, 06-20 00Z 2.573, 06Z 4.174, 12Z 2.492, 18Z 1.042, 06-21 00Z 0.04582, 06Z 1.889, 12Z 3.182, 18Z 3.395, 06-22 00Z 5.284, 06Z 4.083, 12Z 4.048
openaq	Ozone (ozone)	06-19 12Z 0.0236, 18Z 0.02624, 06-20 00Z 0.02021, 06Z 0.01288, 12Z 0.01231, 18Z 0.0135, 06-21 00Z 0.008507, 06Z 0.003192, 12Z 0.02052, 18Z 0.0218, 06-22 00Z 0.01395, 06Z 0.01341, 12Z 0.01388
openaq	PM10 (pm10)	06-19 12Z 50.56, 18Z 45.94, 06-20 00Z 23.06, 06Z 30.39, 12Z 30.44, 18Z 11.61, 06-21 00Z 19.28, 06Z 31.67, 12Z 11.19, 18Z 21.61, 06-22 00Z 34.44, 06Z 36.06, 12Z 55
openaq	PM2.5 (pm25)	06-19 12Z 11.26, 18Z 10.17, 06-20 00Z 6.636, 06Z 5.859, 12Z 5.44, 18Z 3.118, 06-21 00Z 4.108, 06Z 6.943, 12Z 2.632, 18Z 4.409, 06-22 00Z 5.88, 06Z 10.04, 12Z 16.58
openweather	Cloud cover (cloud_cover)	06-19 12Z 60.75, 18Z 68.6, 06-20 00Z 97.41, 06Z 94.23, 12Z 99.5, 18Z 99.61, 06-21 00Z 98.57, 06Z 41.63, 12Z 47.2, 18Z 75.93, 06-22 00Z 10.37, 06Z 7.2, 12Z 13.5
openweather	Relative humidity (humidity)	06-19 12Z 75.12, 18Z 72.73, 06-20 00Z 86.86, 06Z 91.13, 12Z 86.37, 18Z 82.23, 06-21 00Z 93.8, 06Z 95.77, 12Z 90.87, 18Z 74.83, 06-22 00Z 84.23, 06Z 91.17, 12Z 88.93
openweather	Precipitation rate (precipitation)	06-19 12Z 0, 18Z 0.035, 06-20 00Z 0.1752, 06Z 0.43, 12Z 1.097, 18Z 0.7358, 06-21 00Z 0.01667, 06Z 0.1767, 12Z 1.158, 18Z 0.464, 06-22 00Z 0, 06Z 0, 12Z 0
openweather	Surface pressure (pressure)	06-19 12Z 1012, 18Z 1011, 06-20 00Z 1012, 06Z 1012, 12Z 1014, 18Z 1012, 06-21 00Z 1011, 06Z 1011, 12Z 1011, 18Z 1010, 06-22 00Z 1011, 06Z 1012, 12Z 1014
openweather	Air temperature (temperature)	06-19 12Z 31.82, 18Z 32.75, 06-20 00Z 27.75, 06Z 27.41, 12Z 28.55, 18Z 27.55, 06-21 00Z 24.3, 06Z 24.69, 12Z 25.49, 18Z 31.42, 06-22 00Z 28.69, 06Z 27.46, 12Z 28.43
openweather	Wind direction (wind_direction)	06-19 12Z 171.4, 18Z 185.2, 06-20 00Z 135.2, 06Z 164.8, 12Z 115.8, 18Z 161.4, 06-21 00Z 121.6, 06Z 168.9, 12Z 165.7, 18Z 175.7, 06-22 00Z 157.9, 06Z 163.4, 12Z 174.3
openweather	Wind speed (wind_speed)	06-19 12Z 2.363, 18Z 2.305, 06-20 00Z 1.742, 06Z 2.456, 12Z 1.399, 18Z 3.311, 06-21 00Z 1.668, 06Z 1.597, 12Z 3.18, 18Z 3.07, 06-22 00Z 2.593, 06Z 2.518, 12Z 2.921

Source	Metric	Values
purpleair	PM2.5 (pm25)	06-19 12Z 12.27, 18Z 11.76, 06-20 00Z 7.532, 06Z 6.934, 12Z 6.522, 18Z 4.924, 06-21 00Z 5.871, 06Z 8.427, 12Z 5.627, 18Z 5.958, 06-22 00Z 7.425, 06Z 13.79, 12Z 15.59
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	06-19 18Z 1, 06-20 18Z 1, 06-21 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	06-19 18Z 1, 06-20 18Z 1, 06-21 18Z 1
sentinel5p	CO column (s5p_co_column)	06-19 18Z 0.02782, 06-21 18Z 0.0279
sentinel5p	CO granules available (s5p_co_granule_available)	06-19 18Z 1, 06-20 18Z 1, 06-21 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	06-19 18Z 0.0001451, 06-21 18Z 0.000131
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	06-19 18Z 1, 06-20 18Z 1, 06-21 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	06-19 18Z 5.001e-05, 06-21 18Z 2.637e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	06-19 18Z 1, 06-20 18Z 1, 06-21 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	06-19 18Z 1, 06-20 18Z 1, 06-21 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	06-19 18Z -2.9e-05, 06-21 18Z -8.157e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	06-19 18Z 1, 06-20 18Z 1, 06-21 18Z 1
tceq	Carbon monoxide (co)	06-19 12Z 0.2111, 18Z 0.2222, 06-20 00Z 0.2833, 06Z 0.1667, 12Z 0.2, 18Z 0.2944, 06-21 00Z 0.3556, 06Z 0.2611, 12Z 0.2059, 18Z 0.2, 06-22 00Z 0.2222, 06Z 0.2278, 12Z 0.3
tceq	Nitrogen dioxide (no2)	06-19 12Z 3.958, 18Z 4.846, 06-20 00Z 7.707, 06Z 2.27, 12Z 4.406, 18Z 6.483, 06-21 00Z 10.06, 06Z 5.241, 12Z 4.138, 18Z 1.519, 06-22 00Z 2.099, 06Z 2.53, 12Z 4.247
tceq	Sulfur dioxide (so2)	06-19 12Z 0.08889, 18Z 0.005556, 06-20 00Z 0.01111, 06Z -0.1, 12Z -0.06111, 18Z -0.1, 06-21 00Z -0.1273, 06Z -0.5, 12Z 0.4294, 18Z -0.4222, 06-22 00Z -0.4, 06Z -0.4556, 12Z -0.39

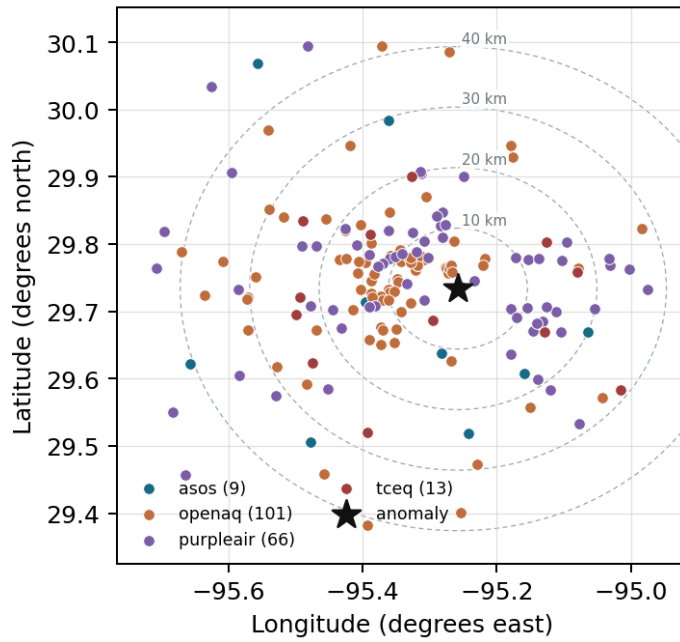
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

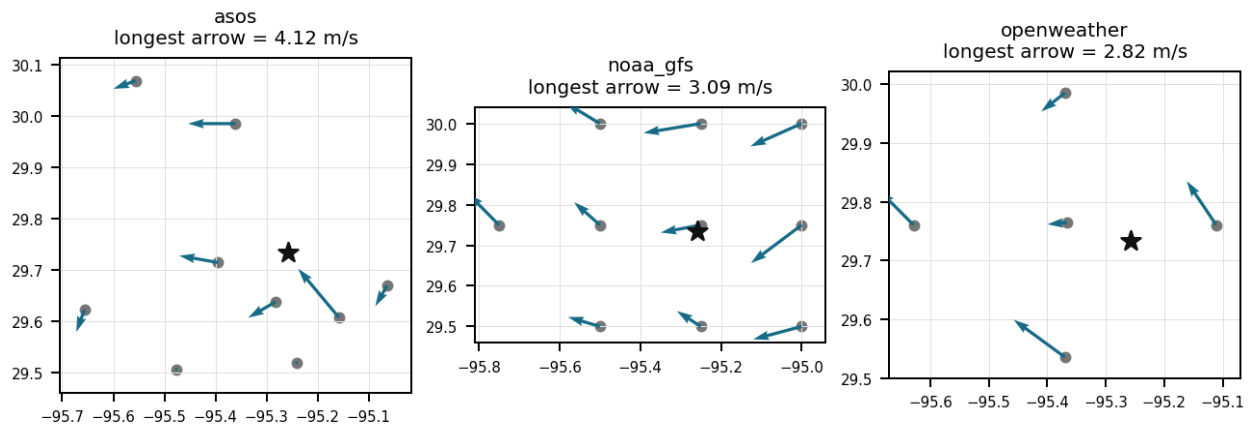
Source	Metric	Values
asos	Relative humidity (humidity)	HOU (10.965 km) 84.75, MCJ (13.448 km) 87.38, EFD (16.998 km) 83.26, T41 (20.02 km) 85.71, LVJ (23.937 km) 84.78, IAH (29.593 km) 87.53, AXH (33.015 km) 81.99, SGR (40.48 km) 86.83, +1 more
asos	Air temperature (temperature)	HOU (10.965 km) 27.85, MCJ (13.448 km) 27.44, EFD (16.998 km) 28.76, T41 (20.02 km) 27.84, LVJ (23.937 km) 27.76, IAH (29.593 km) 27.09, AXH (33.015 km) 27.72, SGR (40.48 km) 27.67, +1 more
asos	Wind direction (wind_direction)	HOU (10.965 km) 136.9, MCJ (13.448 km) 150.1, EFD (16.998 km) 159.7, T41 (20.02 km) 149.7, LVJ (23.937 km) 117.6, IAH (29.593 km) 116.5, AXH (33.015 km) 109.7, SGR (40.48 km) 139.9, +1 more
asos	Wind speed (wind_speed)	HOU (10.965 km) 3.765, MCJ (13.448 km) 4.816, EFD (16.998 km) 4.887, T41 (20.02 km) 3.908, LVJ (23.937 km) 2.601, IAH (29.593 km) 2.951, AXH (33.015 km) 1.801, SGR (40.48 km) 3.401, +1 more
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.25 (1.952 km) 5895, gfs:29.75,-95.50 (23.473 km) 5895, gfs:29.75,-95.00 (24.937 km) 5895, gfs:29.50,-95.25 (26.0 km) 5896, gfs:30.00,-95.25 (29.617 km) 5894, gfs:29.50,-95.50 (34.993 km) 5896, gfs:29.50,-95.00 (35.994 km) 5896, gfs:30.00,-95.50 (37.722 km) 5894, +2 more
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.25 (1.952 km) 572.6, gfs:29.75,-95.50 (23.473 km) 596.5, gfs:29.75,-95.00 (24.937 km) 559.4, gfs:29.50,-95.25 (26.0 km) 520.5, gfs:30.00,-95.25 (29.617 km) 578.4, gfs:29.50,-95.50 (34.993 km) 516.2, gfs:29.50,-95.00 (35.994 km) 618.2, gfs:30.00,-95.50 (37.722 km) 627.4, +2 more

Source	Metric	Values
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.25 (1.952 km) 53.28, gfs:29.75,-95.50 (23.473 km) 52.99, gfs:29.75,-95.00 (24.937 km) 53.56, gfs:29.50,-95.25 (26.0 km) 53.37, gfs:30.00,-95.25 (29.617 km) 52.83, gfs:29.50,-95.50 (34.993 km) 53.21, gfs:29.50,-95.00 (35.994 km) 53.45, gfs:30.00,-95.50 (37.722 km) 52.5, +2 more
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.25 (1.952 km) 1010, gfs:29.75,-95.50 (23.473 km) 1009, gfs:29.75,-95.00 (24.937 km) 1011, gfs:29.50,-95.25 (26.0 km) 1010, gfs:30.00,-95.25 (29.617 km) 1008, gfs:29.50,-95.50 (34.993 km) 1009, gfs:29.50,-95.00 (35.994 km) 1011, gfs:30.00,-95.50 (37.722 km) 1007, +2 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.25 (1.952 km) 19.37, gfs:29.75,-95.50 (23.473 km) 19.43, gfs:29.75,-95.00 (24.937 km) 19.6, gfs:29.50,-95.25 (26.0 km) 19.52, gfs:30.00,-95.25 (29.617 km) 19.44, gfs:29.50,-95.50 (34.993 km) 19.54, gfs:29.50,-95.00 (35.994 km) 19.67, gfs:30.00,-95.50 (37.722 km) 19.45, +2 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.25 (1.952 km) -0.9299, gfs:29.75,-95.50 (23.473 km) -0.8632, gfs:29.75,-95.00 (24.937 km) -1.051, gfs:29.50,-95.25 (26.0 km) -1.162, gfs:30.00,-95.25 (29.617 km) -0.9732, gfs:29.50,-95.50 (34.993 km) -1.103, gfs:29.50,-95.00 (35.994 km) -1.096, gfs:30.00,-95.50 (37.722 km) -0.8057, +2 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.25 (1.952 km) 2.721, gfs:29.75,-95.50 (23.473 km) 2.727, gfs:29.75,-95.00 (24.937 km) 3.115, gfs:29.50,-95.25 (26.0 km) 3.618, gfs:30.00,-95.25 (29.617 km) 2.497, gfs:29.50,-95.50 (34.993 km) 3.168, gfs:29.50,-95.00 (35.994 km) 4.319, gfs:30.00,-95.50 (37.722 km) 2.647, +2 more
openaq	Ozone (ozone)	3378 (0.021 km) 0.01264, 2039 (5.231 km) 0.01387, 325 (6.369 km) 0.019, 320 (12.059 km) 0.01228, 1319333 (14.033 km) 0.01783, 332 (14.34 km) 0.0198, 3214 (14.86 km) 0.01626, 2800 (17.662 km) 0.01463, +8 more
openaq	PM10 (pm10)	13380082 (0.021 km) 30.37, 15852419 (7.017 km) 34.49, 271 (14.34 km) 23.85
openaq	PM2.5 (pm25)	15539682 (0.005 km) 7.998, 3379 (0.021 km) 9.504, 8967123 (0.058 km) 9.175, 8967479 (0.058 km) 7.907, 9489522 (2.89 km) 6.309, 8967065 (2.933 km) 7.28, 9201801 (3.625 km) 5.127, 12178865 (3.713 km) 7.179, +74 more
openweather	Cloud cover (cloud_cover)	grid:center (10.986 km) 64.31, grid:east (14.451 km) 63.46, grid:south (24.535 km) 66.71, grid:north (29.962 km) 67.38, grid:west (35.93 km) 62.75
openweather	Relative humidity (humidity)	grid:center (10.986 km) 85.08, grid:east (14.451 km) 87.76, grid:south (24.535 km) 85.33, grid:north (29.962 km) 86.35, grid:west (35.93 km) 85.22
openweather	Precipitation rate (precipitation)	grid:center (10.986 km) 0.2808, grid:east (14.451 km) 0.3476, grid:south (24.535 km) 0.3368, grid:north (29.962 km) 0.4374, grid:west (35.93 km) 0.3919
openweather	Surface pressure (pressure)	grid:center (10.986 km) 1011, grid:east (14.451 km) 1012, grid:south (24.535 km) 1012, grid:north (29.962 km) 1012, grid:west (35.93 km) 1011
openweather	Air temperature (temperature)	grid:center (10.986 km) 28.15, grid:east (14.451 km) 28.06, grid:south (24.535 km) 28.22, grid:north (29.962 km) 27.75, grid:west (35.93 km) 27.95
openweather	Wind direction (wind_direction)	grid:center (10.986 km) 143.9, grid:east (14.451 km) 164.1, grid:south (24.535 km) 178.1, grid:north (29.962 km) 140.4, grid:west (35.93 km) 160.7
openweather	Wind speed (wind_speed)	grid:center (10.986 km) 2.19, grid:east (14.451 km) 3.529, grid:south (24.535 km) 2.134, grid:north (29.962 km) 2.165, grid:west (35.93 km) 1.863
purpleair	PM2.5 (pm25)	99797 (0.6 km) 7.307, 166435 (2.677 km) 10.57, 6752 (5.276 km) 7.919, 222601 (6.699 km) 4.777, 133994 (7.377 km) 0.8507, 235425 (8.237 km) 7.733, 229853 (8.54 km) 5.953, 127441 (8.731 km) 10.17, +58 more
tceq	Carbon monoxide (co)	482011035 (0.0 km) 0.1414, 482011039 (14.355 km) 0.1606, 482011052 (15.44 km) 0.4268
tceq	Nitrogen dioxide (no2)	482011035 (0.0 km) 6.073, 482010416 (6.378 km) 2.597, 482011039 (14.355 km) 3.852, 482010026 (14.878 km) 7.154, 482011052 (15.44 km) 10.42, 482011015 (17.437 km) 4.906, 482010024 (19.743 km) 1.353, 482011066 (22.737 km) 5.264, +4 more
tceq	Sulfur dioxide (so2)	482011035 (0.0 km) -0.1159, 482011039 (14.355 km) 0.1729, 482010051 (24.236 km) -0.5386

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide

Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.

Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.

Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.

Missing or thin evidence means Unsure, not Invalid.

A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.

Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 34

Nitrogen dioxide (NO₂) reached a concentration of 17.3 ppb at the same monitoring site and time as the carbon monoxide anomaly.

Mark exactly one:

☐ V

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Note (optional):

Claim 2 of 34

Nighttime stagnation and a shallow boundary layer likely amplified near-surface carbon monoxide at the Houston monitor by suppressing dilution and trapping fresh emissions close to the ground.

Mark exactly one:

☐ V

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Note (optional):

Claim 3 of 34

The air quality anomaly in Houston, Texas is characterized by elevated levels of PM2.5 and NO2.

Mark exactly one:

☐ V

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Note (optional):

Claim 4 of 34

Emission transport from sources east to northeast of the Houston monitor was a plausible contributor because local winds were light and predominantly from about 60 to 85 degrees during the anomaly period, which would advect plumes from the ship channel, industrial corridor, port, or nearby traffic corridors toward the site.

Mark exactly one:

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Note (optional):

Claim 5 of 34

A shallow nocturnal planetary boundary layer and low surface wind speeds suppressed atmospheric mixing, causing primary pollutants to accumulate near the surface.

Mark exactly one:

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Note (optional):

Claim 6 of 34

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:

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Note (optional):

Claim 7 of 34

The Houston carbon monoxide anomaly at 2026-06-21T03:00:00+00:00 is most consistent with nighttime trapping of local primary combustion emissions under weak winds and a shallow boundary layer.

Mark exactly one:

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Note (optional):

Claim 8 of 34

Low regional PM2.5 levels from OpenAQ and PurpleAir alongside steady Sentinel-5P total column CO measurements indicate that large-scale wildfire smoke transport or regional pollution events were unlikely.

Mark exactly one:

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Note (optional):

Claim 9 of 34

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:

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Note (optional):

Claim 10 of 34

Meteorological observations near 2026-06-21T03:00:00+00:00 showed very humid and weak-wind conditions near the anomaly location, including 94.16 percent humidity and 2.057776 m/s wind speed from asos and 95 percent humidity and 0.89 m/s wind speed from openweather.

Mark exactly one:

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Note (optional):

Claim 11 of 34

NOAA GFS model outputs and ASOS weather station observations indicated a shallow nocturnal boundary layer height below 200 meters and light surface winds around 2026-06-21T03:00:00Z.

Mark exactly one:

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Note (optional):

Claim 12 of 34

A severe thunderstorm or tornado event has stirred up dust and pollutants from the city's infrastructure, contributing to the air quality anomaly.

Mark exactly one:

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Note (optional):

Claim 13 of 34

The source direction for the Houston carbon monoxide anomaly is most plausibly from the east to northeast of the monitor because nearby surface winds were light and from about 60 to 85 degrees at the anomaly time.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 0.89 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.43149306530293385 m/s; n=360 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:

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Note (optional):

Claim 14 of 34

The anomaly occurred on June 21st at approximately 03:00:20+00:00, with the nearest sensor located 0.6 km away.

Mark exactly one:

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Note (optional):

Claim 15 of 34

Meteorological data from ASOS, OpenWeather, and GFS show very shallow planetary boundary layer heights below 200 meters and weak surface winds under 2 m/s around 03:00 UTC on June 21, 2026, supporting nocturnal atmospheric trapping.

Mark exactly one:

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Note (optional):

Claim 16 of 34

The PM_{2.5} levels are elevated at 8.1 ug/m³, which is above the mean of 8.3517 ug/m³.

Mark exactly one:

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Note (optional):

Claim 17 of 34

The air quality anomaly in Houston, Texas is caused by high levels of SO₂ particles.

Mark exactly one:

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Note (optional):

Claim 18 of 34

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:

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Note (optional):

Claim 19 of 34

The localized accumulation of carbon monoxide in Houston was driven by nocturnal atmospheric trapping of primary combustion emissions.

Mark exactly one:

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Note (optional):

Claim 20 of 34

Nighttime stagnation and shallow mixing strongly support boundary-layer trapping as a cause of the Houston carbon monoxide anomaly because winds near the anomaly were light at about 0.89 to 2.06 m/s, humidity was about 94 to 95 percent, and GFS planetary boundary layer height was only about 212 m near 00Z and about 140 to 187 m in the surrounding 6-hour period.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 21 of 34

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, leading to an accumulation of pollutants in the atmosphere.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 22 of 34

The pollutant pattern at the Houston monitor at 2026-06-21T03:00:00+00:00 shows fresh combustion influence because carbon monoxide was elevated, nitrogen dioxide was elevated to 17.3 ppb, and ozone was 0.0 ppm.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 23 of 34

At 2026-06-21T03:00:00+00:00, carbon monoxide (CO) reached 0.4 ppm at coordinates (29.734, -95.258), significantly above the expected concentration of 0.128 ppm with a z-score of 5.10.

Mark exactly one:

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Note (optional):

Claim 24 of 34

The temperature on June 21, 2026 was relatively low, with a mean value of 24.16°C.

Mark exactly one:

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Note (optional):

Claim 25 of 34

Regional wildfire smoke or a sulfur-rich industrial plume is not strongly supported as the main cause of the Houston carbon monoxide anomaly because nearby PM2.5 was only about 6.6 to 8.1 ug/m3, PM10 was about 21 ug/m3, sulfur dioxide at the colocated monitor was about -0.1 ppb, and the later Sentinel-5P sulfur dioxide column was not elevated.

Mark exactly one:

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Note (optional):

Claim 26 of 34

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:

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Note (optional):

Claim 27 of 34

The tceq carbon monoxide measurement at 29.734, -95.258 was 0.4 ppm at 2026-06-21T03:00:00+00:00, compared with an expected value of 0.12760736196319017 ppm, corresponding to a z-score of 5.095711951734264 and a severe anomaly classification.

Mark exactly one:

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Note (optional):

Claim 28 of 34

The TCEQ monitoring station in Houston recorded a carbon monoxide concentration of 0.4 ppm and a nitrogen dioxide concentration of 17.3 ppb at 2026-06-21T03:00:00Z.

Mark exactly one:

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Note (optional):

Claim 29 of 34

Fresh local combustion emissions likely contributed to the carbon monoxide anomaly because carbon monoxide and nitrogen dioxide were elevated while ozone was near zero, a pattern consistent with recent primary emissions under nocturnal conditions.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 30 of 34

The wind direction and speed were not significantly different from the average values for the region.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 31 of 34

TCEQ ground monitoring recorded a simultaneous increase in nitrogen dioxide concentrations to 17.3 ppb alongside carbon monoxide at 03:00 UTC on June 21, 2026, supporting a localized primary combustion source hypothesis.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 32 of 34

Localized combustion emissions from vehicular traffic or industrial sources near the monitoring site produced coincident elevated concentrations of carbon monoxide and nitrogen dioxide.

Mark exactly one:

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Note (optional):

Claim 33 of 34

The nearest openaq ozone measurement at the anomaly location was 0.0 ppm at 2026-06-21T03:00:00+00:00, while the nearest tceq nitrogen dioxide measurement at the same location and time was 17.3 ppb.

Mark exactly one:

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Note (optional):

Claim 34 of 34

Fresh local combustion supports the Houston carbon monoxide anomaly because the anomaly coincided with elevated local nitrogen dioxide at 17.3 ppb and very low ozone near 0.0 ppm, which is consistent with recently emitted combustion plumes under nighttime ozone titration rather than daytime secondary production.

Mark exactly one:

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Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.
